Brainy boffs make quantum computers!
Engineers at Yale have devised a process to create a silicon chip that contains all the components needed for a quantum information processor. **http://dgit.in/quantum_computerx**

Gaming sneakers!
Nintendo and Vans team up to start a chain of retro gaming sneakers.
http://dgit.in/gaming_sneakers

19th century and the early 20th century, there were a couple of efforts to build gas and steam powered movement assisters. But these weren't really eligible for being classified as exoskeletons. The first true mobile machine which integrated with human movements was developed almost 50 years later by General Electric and the US Military in the 1960s. The suit was named Hardiman, and it enabled the user to lift weights up to 680kgs. It effectively reduced weights to their 25th portion, i.e., a person lifting 25kg would feel like the weight is 1kg. Ultimately, the size and weight (almost 1500kgs) with the power supply and other control issues never let Hardiman go beyond the prototype stage, and it was declared too unsafe to use. A number of further projects did spring up over the next 40 years. The Pitman suit designed by the Los Alamos National Laboratory scientists never left the drawing board. Even in the 90s, efforts by the US Army to build an iron-man like suit met the same fate. The limitations of the technology available had stymied the possibilities of exoskeletons for the entire duration of the last century.

Fast forward to 2000, the Defence Advanced Research Projects Agency (DARPA) kickstarted a number of human exoskeleton projects under the banner of "Exoskeletons for Human Performance Augmentation" project, with over $75million in funds. Their requirements defined a suit that would let a soldier carry a

The Solotrek XFV is the exoskeleton that can fly

TAKE TO THE SKIES!

The Solo Trek XFV (eXoskeletor Flying Vehicle) is actually a work in progress flying exoskeleton that has promised a range of 125 miles, vertical takeoff and landing, hovering ability at 8000 metres and a cruise speed of almost 130kmph.

heavy load for days without tiring, be able to operate heavy weapons single-handedly, to jump significantly high and be immune to enemy fire. Or one could say, they wanted Iron Man.

Needless to say, there were not many takers for the adventurous proposal. And from the ones that did take it up, only few have stayed in the project so far. Out of these, a few of the noteworthy ones were the XOS series from Raytheon, Lockheed Martin's HULC and Millennium Jet's Solo Trek XFV. In 2010, Raytheon, an international defense contractor demonstrated XOS2, a wearable body armor guided by muscle contractions, that could lift almost thrice the weight as a normal human being. The XOS series was actually the brainchild of a company called Sarcos, which even went on to get approval from the military for their design, before it was taken over by Raytheon which continued its original undertakings. Their original design detected the contraction of human muscles and in turn pumped high pressure hydraulic fluid to joints on the setup, which in turn moved cylinder actuators attached to cables that acted as tendons for the mechanical muscles. This was closest to what the military had envisioned.

HULC, which had its origins in the Berkeley Lower Extremity Exoskeleton (BLEEX) from UC Berkeley's Robotics and Human Engineering Laboratory, has been developed to be powered by fuel cell power to make it a completely untethered military exoskeleton. Berkeley had already been working on reducing the energy consumption from the limbs of the suit and it did so with some success, since the Human Load Carrier (HuLC) could reportedly work for 20 hours without recharging.

Apart from all these military advancements, exoskeletons can also be put to use for medical purposes. People with spinal injuries or partial disabilities could get around with as much ease as fully abled people with powered exoskeletons enabling them to do what their own bodies can't. In fact, initial versions of such devices are already in the market. Take Argo Medical Technologies' Rewalk device, a $150,000 exoskeleton that detects the shift of weight from the wearer and uses the motors at the hip and knee joints to essentially let them walk. It is actually in its sixth generation now and is claimed by the creators to be the most researched exoskeleton.

Around the same time as XOS2, Japan's Cyberdyne Inc. (yes, we hear you screaming "Judgement day") developed

A man wearing the Cyberdyne Hybrid Assistive Limb suit (left)

HAL (Hybrid Assistive Limb), something that could finally be called a robotic suit. Contrary to the muscle contraction detection seen in the XOS series, this model has been developed to detect brain signals which would otherwise be meant for corresponding muscles only. Theoretically, this meant that the wearer only needed to think about moving to move. The exoskeleton itself is made up of a combination of sensors, actuators and controllers. This suit was actually used by relief workers after the Fukushima nuclear plant meltdown and can be purchased by Japanese hospitals for qualified patients for $20,000.

Also in the medical exoskeleton domain is the Phoenix exoskeleton